

Interpretation

- K = “heat kernel”/smooth cost
- $\mathbf{u}, \mathbf{v}$ = balancing multipliers

EOT solution has a nice form

One can show that:

$$\pi^\varepsilon = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v})$$

$$\pi_{ij}^\varepsilon = \mathbf{u}_i \mathbf{K}_{ij} \mathbf{v}_j$$

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For some scaling vectors $\mathbf{u}, \mathbf{v}$, and for $\mathbf{K}_{ij} = \exp\{-\mathbf{C}_{ij}/\varepsilon\}$

The scalings have to be such that $\pi^\varepsilon \mathbf{1} = \mathbf{a}$, $(\pi^\varepsilon)^\top \mathbf{1} = \mathbf{b}$

This suggests an algorithmic approach!

Sinkhorn's Algorithm